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Savitribai Phule Pune University, Pune Maharashtra, India



A PROJECT ON
“CREDIT CARD
FRAUD DETECTION”
AT
TECHNOSIGNIA
SUBMITTED TO
SAVITRIBAI PHULE PUNE UNIVERSITY

IN PARTIAL FULFILLMENT OF
MASTER OF SCIENCE IN DATA SCIENCE

SUBMITTED BY
AVINASH PADGHAN
(2025-26)
UNDER THE GUIDANCE OF
MR. SUYASH PATANKAR



AAER's Asian College of Science and Commerce
PARI Company Chowk, Dhayari-Narhe Rd, Tal-Haveli, Dhayari, Pune,
Maharashtra 411041

CERIFICATE

This is to certify that, **AVINASH SAMADHAN PADGHAN**, a student of Master of Science in Data Science with Roll Number_____ has successfully completed the **On-the-Job Training (OJT) DISSERTATION** titled **CREDIT CARD FRAUD DETECTION** under the guidance and supervision of **MR SUYESH PATANKAR**, during the academic year **2025-26**.

The DISSERTATION was of exceptional quality, demonstrating a high level of technical proficiency, creativity, and innovation, and has been recognized as an outstanding contribution to the field of DATA ANALYST. It was evaluated and found to exceed the academic standards set by the Department of Computer Science, **Asian College Of Science And Commerce**, and the affiliated **Savitribai Phule Pune University, Pune**. making it a remarkable achievement in the B curriculum.

We commend the student's exemplary effort, dedication, and outstanding performance throughout the DISSERTATION.

Guide

Dept. of Computer Science

H.O.D

Dept. of Computer Science

Principal

External

DECLARATION

I am **AVINASH SAMADHAN PADGHAN**, Roll Number____, a student of **Master of Science in Data Science**, hereby declare that the DISSERTATION titled “**CREDIT CARD FRAUD DETECTION** ” is a result of my own work, undertaken as part of the requirements for the award of the **MSC-(DS)** degree at **Asian College Of Science And Commerce Pune**, and the affiliated **Savitribai Phule Pune University, Pune**. This DISSERTATION has not been submitted for any other degree or award elsewhere. All the information, data, and findings presented in this DISSERTATION are authentic to the best of my knowledge, and due acknowledgment has been made for any references used from books, journals, websites, or any other sources.

I further declare that this work is the result of my sincere efforts, and the assistance provided by **MR SUYESH PATANKAR**, my DISSERTATION guide, has been duly acknowledged.

I take full responsibility for the content and conclusions drawn in this DISSERTATION

Place: Asian College Of Science And Commerce Pune

Mr. MR SUYESH PATANKAR

PROFILE



Name Of Employee: AVINASH SAMADHAN PADGHAN

Name Of Project : Credit Card Fraud Detection Using Machine Learning

Name Of Client : TechnoSignia

Role Of Employee : Data Science Intern

Technologies : Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn,
Jupyter Notebook

Education : M.Sc. (Data Science)

University : Savitribai Phule Pune University (SPPU), Pune

Project Type : Machine Learning / Classification

Domain : Banking / Financial Service

Key Activities : Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Feature Scaling, Train-Test Split, Data Imbalance Handling, Machine Learning Model Building, Model Evaluation

A **Data Science Intern** works with data to understand patterns, prepare datasets, build Machine Learning models, and evaluate their performance. The role includes activities such as data cleaning, data preprocessing, exploratory data analysis, feature preparation, model building, and result analysis using Python and Machine Learning libraries.

In the **Credit Card Fraud Detection** project, the Data Science Intern analyses credit card transaction data to identify genuine and fraudulent transactions. The project involves cleaning and preprocessing the dataset, performing Exploratory Data Analysis (EDA), analysing transaction amounts and fraud distribution, handling data imbalance, and applying feature scaling and train-test splitting.

Different Machine Learning classification algorithms such as **Logistic Regression, Decision Tree, Random Forest Classifier, and K-Nearest Neighbors (KNN)** are implemented and compared. The models are evaluated using performance metrics to understand their effectiveness in detecting fraudulent transactions. Based on the model comparison, **Random Forest Classifier achieved the highest accuracy of 96.97%** among the four models tested in the project.

MR. AVINASH SAMADHAN PADGHAN

